Some proofs

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The kaTex rendering for `.md` files is horrendus, thus here is a nicer LaTeX format

1 Simple Linear Regresion model

We define our SLR model.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \tag{1}$$

Where Y_i is our response variable, β_1 and β_0 is both our parameters and ϵ_i is the random error term. We can start off with our assummptions for SLR, which are $\epsilon_i \overset{i.i.d}{\sim} N(0, \sigma^2)$. Since ϵ_i is a random variable with an assumed Gaussian distribution, It is obvious to see that Y_i is then a random variable from the Gaussian distribution. Such that

$$Y \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \tag{1.2}$$

To find this is trival, we take the expected value of Y_i such that

$$\mathbb{E}[Y_i] = \mathbb{E}[\beta_0 + \beta_1 X_i + \epsilon_i] = \mathbb{E}[\beta_0] + \mathbb{E}[\beta_1 X_i] + \mathbb{E}[\epsilon_i] = \beta_0 + \beta_1 X_i$$

$$\mathbb{E}[Y_i] = \beta_0 + \beta_1 X_i \tag{1.3}$$

X_i is our known constant, β_0 and β_1 are the unknown constants, and ϵ_i is assumed to have $\mathbb{E}[\epsilon_i] = 0$.

1.1 Least Squares method

We are sometimes interested in the observed values deviation from its mean to distinguish the variabillity within the distribution of Y_i. Let Q define the least squares equation for Y_i.

$$Q = \sum_{i=1}^{n} (Y_i - \mathbb{E}[Y_i])^2 \tag{1.4.1}$$

A high Q indicates high variability within the observed values, this is a cause of problems for when building a model to draw predictions. We can minimize Q w.r.t the regression model parameters. In order to do this we borrow the equation from (1.3) and set the partial derivative equal to zero.

$$\frac{\partial Q}{\partial \beta_1} = -2 \sum_{i=1}^{n} X_i (Y_i - \beta_0 - \beta_1 X_i) = 0 \tag{1.5.1}$$

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum_{i=1}^{n} (Y_i - \beta_0 - \beta_1 X_i) = 0 \tag{1.5.2}$$

We apply the chain rule. From the above equations, we can derive the normal equations and the linear estimators b_1 and b_0. We will conceal the index of the sums from here on out. Below is the first normal equation.

$$-2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i) = 0$$

$$\sum (Y_i X_i - b_0 X_i - b_1 X_i^2) = 0$$

$$\sum Y_i X_i - b_0 \sum X_i - b_1 \sum X_i^2 = 0$$

$$\sum Y_i X_i = b_0 \sum X_i - b_1 \sum X_i^2 \qquad (1.6.1)$$

$$
\begin{aligned}
-2\sum(Y_i - \beta_0 - \beta_1 X_i) &= 0 \\
\sum(Y_i - b_0 - b_1 X_i) &= 0 \\
\sum Y_i - nb_0 - b_1 \sum X_i &= 0
\end{aligned}
$$

$$
\sum Y_i = nb_0 - b_1 \sum X_i \qquad (1.6.2)
$$

(1.6.1) (1.6.2) are the derived normal equations.

2 Least Square Estimators